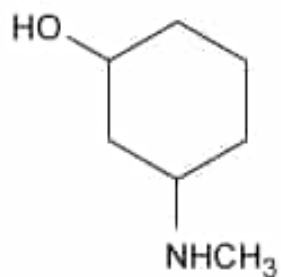


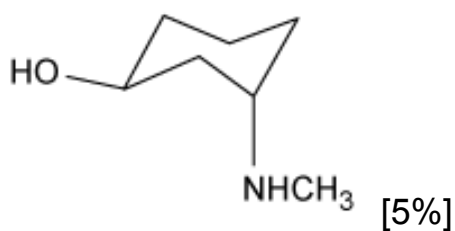
Compound **1** is shown below.



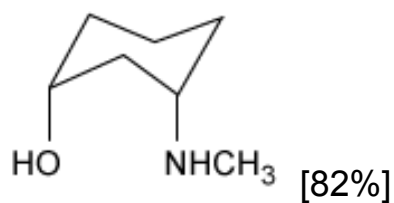
Compound **1**

Which of the following conformations of Compound **1** has intramolecular hydrogen bonding?

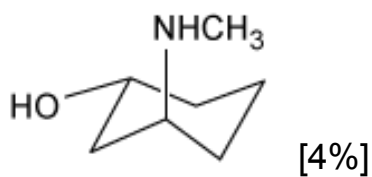
☐ A.



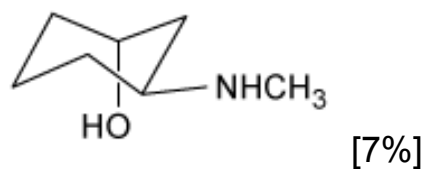
✓ ☒ B.



☐ C.



☐ D.



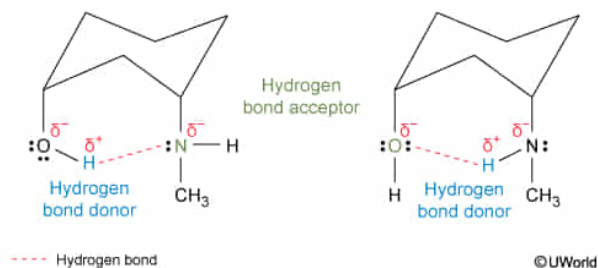
Correct

83% answered correctly

Explanation:

Hydrogen bonding

—NH₂ and —OH are in axial conformation and can hydrogen bond, which stabilizes the compound.



Hydrogen bonding is an intermolecular or intramolecular force that occurs between a hydrogen bond **donor and acceptor**. Hydrogen bond donors are hydrogen atoms bonded to a second-row electronegative atom. Hydrogen bond acceptors are electronegative atoms with a lone pair of electrons, typically oxygen, nitrogen, or fluorine. The electronegative atom pulls electrons toward itself and away from the hydrogen atom, creating a **dipole**. The dipole causes a partial negative charge (δ^-) on the electronegative atom and a partial positive charge (δ^+) on hydrogen. Hydrogen bonds can **help stabilize** a molecule.

Compound **1** contains an **alcohol** and an **amine**, both of which can participate in hydrogen bonding. The hydroxyl oxygen and the amine nitrogen both have partial negative charges and are hydrogen bond acceptors. The hydroxyl and amine hydrogens have partial positive charges and are hydrogen bond donors. When both functional groups are in the **axial position**, they align and can hydrogen bond with one another (intramolecular), stabilizing the molecule (**Choice B**).

(Choices A, C, and D) Each of these conformations has one axial substituent and one equatorial substituent. Therefore, the substituents are not close in proximity and cannot hydrogen bond with each other.

Educational objective:

Hydrogen bonding is an intermolecular or intramolecular force that occurs between a hydrogen bond donor and acceptor; this type of force helps stabilize molecules. Hydrogen bond acceptors are electronegative atoms with a lone pair of electrons (eg, oxygen and nitrogen), and hydrogen bond donors are hydrogen atoms bonded to an electronegative atom. Alcohols and amines contain groups that can act as both hydrogen bond donors and acceptors.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 402550

System:

Functional Groups and Their Reactions